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Storage Services

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Content

- Object Storage
- Block Storage
- File Storage
- AWS Storage Services:
- Various Storage Classes of S3
- Amazon EFS
- Amazon EBS
- Create an amazon EBS volume
- Hosting a Static Website using S3

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Storage Types in Cloud

- Object Storage – Stores data as objects (Amazon S3)
- Block Storage – Stores data in fixed-size blocks (Amazon EBS)
- File Storage – Hierarchical file system (Amazon EFS)

AWS Storage Services Overview

- Amazon S3 – Scalable object storage
- Amazon EBS – Block-level storage for EC2
- Amazon EFS – Elastic File Storage
- Amazon Glacier – Archival storage

Amazon S3 (Object Storage)

- Stores data as objects in buckets
- Virtually unlimited storage
- Use cases: Backup, media hosting, static websites
- Durability (11 9’s), scalability, security

S3 Storage Classes

- S3 Standard – Frequently accessed data
- S3 Intelligent-Tiering – Auto tiering
- S3 Standard-IA – Infrequent access
- S3 One Zone-IA – Single AZ, low cost
- S3 Glacier – Archival storage
- S3 Glacier Deep Archive – Cheapest, long-term

Amazon EFS (File Storage)

- Managed, scalable file storage
- Supports NFS protocol
- Shared access across EC2
- Use cases: Content management, DevOps, big data

Amazon EBS (Block Storage)

- Attachable block-level storage for EC2
- Persists after instance stop/terminate (if configured)
- Types: gp2/gp3, io1/io2, st1, sc1

Creating an EBS Volume

- 1. Open EC2 Console → Volumes → Create Volume
- 2. Select volume type, size, availability zone
- 3. Attach volume to EC2 instance
- 4. Login via SSH → Format & Mount volume

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Hosting a Static Website on S3

- 1. Create an S3 Bucket
- 2. Upload HTML, CSS, JS files
- 3. Enable Static Website Hosting
- 4. Set bucket policy for public read access
- 5. Access via S3 website endpoint

Summary

- Object vs Block vs File Storage
- AWS Storage Services: S3, EFS, EBS
- S3 Classes for cost optimization
- Hands-on: EBS creation & S3 website hosting

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